Overtraining Syndrome in Young Athletes: Causes, Physiological and Psychological Consequences, and Evidence-Based Prevention Strategies

by

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Abstract

Overtraining syndrome (OTS) is a maladaptive condition resulting from a prolonged imbalance between training stress and recovery, leading to a persistent decrease in athletic performance, chronic fatigue, and a wide spectrum of physiological, psychological, and behavioral disturbances. Although OTS has been extensively studied in professional and elite adult athletes, its manifestation in young athletes remains comparatively less studied. Youth athletes are particularly vulnerable due to their ongoing physical growth, hormonal development, academic demands, social expectations, and pressure to achieve competitive success at increasingly young ages. This paper explores the causes, physiological and psychological symptoms, and prevention strategies of OTS in young athletes through a meta-analysis of the scientific literature. This paper is intended to identify the major risk factors and propose healthier approaches to youth athletic development.

Keywords: overtraining syndrome, adolescent athletes, physiological and psychological symptoms, prevention strategies

1. Introduction

1.1. Background

Physical training is considered essential for enhancing physical performance and promoting general health. Contemporary society actively promotes exercise as an effective means of promoting both physical and psychological well-being. Moderate physical activity is beneficial in ways that enhance immunity, reduce psychological stress, and the like. However, the effects of training are not purely black-and-white. Indeed, symptoms such as excessive fatigue and persistent muscle soreness are among the most prevalent in athletes. While those causal signs are often dismissed as natural consequences of training, they could be a potential sign of chronic burnout.

Physical training is considered inseparable from good performance, as performance depends heavily on the time they devote to perfecting their skills. For this reason, young athletes, in particular, are more prone to overtraining syndrome. This is explained by the cultural perspective of athletic careers, in which young athletes are encouraged to be pushed to their physical limits during their prime years. In turn, overtraining syndrome may be permanently detrimental to their future performances. However, it is misleading to treat training as the sole key to good performance. Excessive training, in fact, often entails physical and mental fatigue and, in the long term, can lead to a permanent drop in performance. This phenomenon is referred to as overtraining syndrome (OTS). OTS occurs when training is excessive to the point that the body cannot manage the physical stress, leading to long-term failure to recover. Its slow healing process characterizes OTS.

1.2. Objectives & Significance

This research aims to investigate the causes and symptoms of these conditions and their association with the human body and psyche. Overall, this study seeks to distinguish OTS from functional overreaching and other fatigue-driven malfunctions and suggest possible strategies to prevent and manage this syndrome.

Research on overtraining syndrome is essential, especially given the vague diagnostic criteria. In this increasingly dynamic sports industry, understanding overtraining syndrome, which predominantly affects elite athletes, will become crucial. The understanding includes the condition's psychological and physiological associations, as well as effective methods for its prevention and management. Understanding the syndrome will offer a new perspective on improving athletic performance, especially for athletes who are experiencing prolonged burnout and unexplained symptoms.

2. Hypothesis

Athletes who are exposed to overtraining or overreaching are at significantly greater risk of damage due to the additional physiological and psychological demands. Also, overtraining syndrome can be prevented with prompt remedial action.

3. Literature Review

3.1. Concepts

Overtraining Syndrome

Overtraining syndrome (OTS) is a condition that includes decreased performance and fatigue, predominantly affecting elite athletes. Metabolic, immune, hormonal, and other dysfunctions result from an imbalance between training stress and adequate recovery. It also

involves changes in physiological functions, impairments in psychological processing, and biochemical abnormalities. Psychological, endocrine, physiological, and immunological factors all contribute to the failure to recover from exercise. Careful monitoring of athletes and their response to training may help prevent the breakout of overtraining syndrome. Reflecting its multisystematic nature, the diagnosis of OTS can be based on specific hormones, neurotransmitters, metabolites, and psychological, electrocardiographic, and electroencephalographic patterns. Diagnosis of OTS may require a combination of these variables.

Causes of Overtraining Syndrome

The extreme standards of competitive sports have rarely been detrimental to athletes involved in highly demanding training. Also, changes in aesthetic standards have led individuals to seek physical exercise to reduce body mass, increase muscular mass, and improve aerobic fitness. It is quite common for athletes and non-athletes alike to exceed the limits of their physical and psychological capacities, triggering overtraining syndrome. In other words, OTS is often caused by excessive training and insufficient rest and recovery. Also, it can be triggered solely by non-training factors, such as eating patterns, regardless of training patterns. Moreover, OTS is characterized by dysregulation of the central nervous, autonomic, and neuroendocrine systems, resulting in physical symptoms and behavioral changes.

Overtraining Syndrome Symptoms

The symptoms of OTS include changes in mood, such as depressed mood, or behavior, decreases or increases in the concentrations of various blood molecules, and alterations in immune function. Moreover, it leads to central nervous system (CNS) effects, including further inflammation, central fatigue, and ultimately neurohormonal changes. In addition, decreased

performance, loss of motivation, emotional instability, and increased incidence of injuries are also included in the OTS symptoms.

3.2. Developments

Recent research in sports science has focused on improving methods for detecting and preventing overtraining syndrome. One important development in the treatment of OTS is the recognition of the importance of early detection. Also, it is found that a well-balanced training program that gradually increases workloads in a logical progression should be used for prevention.

3.3. Research Gap

Although overtraining syndrome has been widely studied in professional and elite athletes, relatively few studies focus specifically on young athletes. Adolescent athletes are likely to experience a range of physiological and psychological responses because of ongoing physical and psychological development. Therefore, additional research focusing on adolescent athletes is essential to better understand the mechanisms, risk factors, and prevention strategies associated with overtraining syndrome.

4. Methodology

The methodology of this study would be a meta-analysis based on a literature review. The sources for this study will primarily be drawn from Google Scholar, a credible platform that provides access to a wide range of peer-reviewed research papers. The investigation will

primarily be based on keywords such as “overtraining syndrome,” “functional overreaching,” and “excessive training effects,” which are necessary to define the study's focus and scope.

5. Data & Results

Causes of Overtraining Syndrome

The causes of overtraining syndrome vary, given its abstract nature and limited research. In a study on this topic, we observed common ground between the physiological and psychological causes of overtraining syndrome: both are associated with insufficient recovery and loss of homeostasis. For example, physiological symptoms largely stem from severe weight loss from excessive training. Similarly, psychological symptoms are attributed to a prolonged state of over-arousal. “Chronic imbalance” and “failed recovery” were key to our findings. We noted that the daily rhythms of athletes are significantly distinct from those of the vast majority.

Physiological Symptoms of Overtraining Syndrome

The association between bone mass and overtraining poses a threat to adolescent athletes' future performance. Adolescence is characterized by rapid physical maturation, and this paper specifically discusses skeletal growth. Note that over 90% of bone mass is achieved by the age of 18, and that failure in bone accrual during this period is highly associated with future risk of fracture. Normally, weight-bearing exercise is associated with higher Bone Mineral Density (BMD). However, for athletes in particular, BMD was found to be lower than the average. Its causes are indirectly related to overtraining, with an imbalance in food intake and excessive energy expenditure resulting in hormonal alterations.

Of importance, BMD is particularly associated with reproductive hormones, especially estrogen. Sufficient estrogen production links to inhibiting bone resorption, supporting rapid

bone mass accrual. When excessive training is coupled with insufficient food intake, the hypothalamic-pituitary-gonadal (H-P-G) axis functions abnormally, failing to promote bone repair and growth. Another common symptom of overtraining syndrome, amenorrhea(absence of menstrual period), was found to occur with a decrease in BMD simultaneously. A similar decrease in fat mass, energy deficit, and malfunctioning H-P-G axis is observed in females with anorexia nervosa. It is known that overtrained athletes consume less carbohydrate per kilogram of body mass per day than non-overtrained athletes. This demonstrates, significantly, that immunosuppressive marker levels and hormonal disruption are higher than in those meeting recommended carbohydrate intakes. Micronutrient deficiencies are prevalent among overtrained athletes and are independently associated with immune suppression and hormonal disruption.

This suggests the fat's role as a hormone regulator or its impact on the H-P-G axis. Collectively, the overlapping symptoms of lower BMD and amenorrhea are both explained by the imbalance of food input and calorie-consuming activity, reducing the role of fat as a controlling hormone essential to proper functioning. In conclusion, training to an excessive degree, especially in the stage of skeletal development, is fatal for adolescent athletes as it increases vulnerability to fractures and hormonal diseases, ultimately making them short-lived as athletes.

Still, the uniquely consistent biomarkers of OTS remain absent, as the researchers found that the athletes had altered combinations of markers each time. Contrary to traditional belief, dietary factors are more significant than excessive training itself. Its damage is maximized when detrimental eating patterns are coupled with excessive energy output. Furthermore, deficits in carbohydrate, protein, and overall caloric intake can independently cause overtraining syndrome. Because each nutrient is responsible for distinct physical functions, an individual can develop

OTS without other risk factors. Still, OTS is not confined to a single trigger but arises from combinations of OTS triggers.

Psychological Symptoms of Overtraining Syndrome

As overtraining progressed toward greater chronicity, the Profile of Mood States, which assesses emotional shifts in tension, depression, anger, vigor, fatigue, and confusion, showed that positive moods declined while negative moods increased. Also, it is known that an overtrained athlete may exhibit emotional and anger, hostility, and motivational imbalance, and mood swings are common. Loss of self-confidence, irritability, excessive, prolonged weariness, fatigue, depression, anxiety, and confusion are the other common psychological symptoms of OTS.

Eating disorders, such as anorexia nervosa and generalized loss of appetite, are observed. Overtraining is associated with body weight loss, while chronic negative energy balance is an inadequacy in food intake that individuals adapt to, defined as a weight-stable energy balance state with low body weight and fat stores in the presence of lower-than-normal energy intake. Many athletes, especially female athletes, are chronically energy deficient. Due to the energy cost of volume and high intensity, prolonged exercise is significant. It has been proposed that excessive exercise and overtraining could result in chronic negative energy balance. When this chronic energy deficit is intentional, it is related to anorexia nervosa, which is more frequent among female athletes. The prevalence of hyperactivity, excessive exercise, and anorexia nervosa lies between 31% and 80%, depending on the study and its criteria for hyperactivity. The existence of exercise-induced suppression of appetite and resistance to begin eating has been indicated, although more research in this area is warranted.

Another psychological symptom of OTS is a change in sleeping patterns. If an athlete gets insufficient sleep, the body is denied its optimal recovery time, which would change the

body's ability to adapt to the stress of training. For example, early-morning training and competition times have been shown to reduce athletes' sleep duration and increase pre-training fatigue. Moreover, in a study of Australian rules football players, evening games were associated with later sleep onset, shorter time in bed, and less total sleep than day games. Furthermore, one study found that a 16-year-old female experiencing overtraining reported several sleep disorders, such as insomnia, restless legs syndrome, and delayed sleep phase. Insufficient sleep, due to the sleep orders, not only impacts attention and learning but also increases the risk of accidents, hypertension, diabetes, and depression. It is even linked to a higher risk of self-harm and suicidal ideation for teenagers.

Impaired cognitive processes have been reported in cases of overtraining. The signs include difficulty concentrating at work and during training, increased internal and external distractibility, decreased ability to handle large amounts of information, decreased ability to narrow focus, and decreased self-esteem. One study used the Stroop Color Word Test, which assesses selective attention and cognitive flexibility, to evaluate a population of clinically diagnosed OTS athletes. Response time on the Stroop task increased in athletes who had been diagnosed as overtrained. Also, the number of mistakes the participants made, or the error rate, increased as training overload increased or was present. This indicates that overtraining can impair cognitive processing.

Prevention Strategies

One prevention strategy is to use the session rate of perceived exertion (sRPE) multiplied by the workout duration as the training load. The sum of the session workload over a week or day represents the training load, while variance can be analyzed by day or week to give a measure of monotony. Many studies on overtraining rely on earlier measures of training load,

monotony, and strain, thereby further supporting the use of a training log with session RPE. It subjectively quantifies an individual's perception of an activity's physical demands. It is a sensitive tool that may detect accumulated fatigue across multiple training days and serve as a surrogate marker of exercise intensity. By using sRPE, coaches, health scientists, and practitioners may gain insight into accumulated fatigue during periods of increased training, helping them avoid inadequate recovery or overtraining. Therefore, it helps individuals learn to self-monitor physical exertion and enhances exercise adherence.

Moreover, psychomotor speed tests could prevent OTS. Psychomotor speed could provide predictive value for athletes at risk of overreaching. It is the neurological ability to receive sensory stimuli from the environment, process them mentally, and eventually perform coordinated actions with large muscles within a certain period of time. Young athletes who engage in long-term physical activity at moderate to high intensity exhibit higher psychomotor performance. By using psychomotor speed, they can track whether they are overtraining or overreaching.

Tapering could also prevent OTS. Tapering is a reduction in training load over a variable period; therefore, it can reduce physiological and psychological stress and optimize sports performance. There are several types of tapering, including nonlinear, exponential, and step tapering. Nonlinear tapering techniques have a more pronounced positive impact on performance than step taper strategies. The primary aim of tapering should be to minimize accumulated fatigue, not to attain additional physiological adaptations or fitness gains. Tapering, when applied in conjunction with pre-taper overload training, is more effective at maximizing performance gains. One study suggests a taper lasting up to 21 days, in which training volume is progressively reduced by 41–60% while training intensity and frequency remain unchanged. Even with

tapering, maintaining training intensity is necessary to avoid detraining, provided that reductions in other training variables allow sufficient recovery to optimize performance.

Periodization is a systematic progression: a resistance training program that follows predictable patterns of change in training variables by using different segments of a microcycle, mesocycle, or macrocycle, with the appropriate manipulation and sequencing of volume. Periodization is a generally accepted approach to managing athletic performance by subdividing training programs into sequential, focused training periods; therefore, it helps prevent athletes from developing overtraining syndrome. Periodization introduces structured variability into the training process to maximize performance in the most critical phases of the competitive season and to improve long-term development.

6. Analysis & Discussion

6.1. Discussion

Our research findings summed up to align with the hypothesis: athletes were indeed more vulnerable to both physical and psychological damage after experiencing overtraining syndrome. Physiological causes, in a bigger scope, stemmed from severe weight loss and nutrition deficit, not only limited to overtraining syndrome. Psychological causes were largely a failure in adapting to the severely depleting condition, ultimately losing homeostasis. Adolescents, in particular, were at higher risk of overtraining syndrome, considering their underdeveloped psyche and body. We have noted that the under-19 age group is the most critical period for skeletal development and accrual. Deficiency of any core nutrient during this period is fatal: not only does it cause temporary fatigue, but it also poses a high risk of underdevelopment, leading to permanent damage. In addition, their adverse condition, in which they simultaneously play

both roles of a student and an athlete, exacerbates sleep deficiency. In turn, they are exposed to a greater risk of hormonal dysfunction and imbalanced eating habits, creating a vicious cycle where mental vulnerability leads to physical malfunctioning. As we note that overtraining syndrome can be physically fatal for adolescents, it will become increasingly important to provide them with proper clinical advice and education, including its warning signs: persistent fatigue and declines in performance. Still, clinical treatment methods remain largely absent, underscoring the need for future research. Given the wide range of causes of OTS, a treatment employing multiple methods is ideal. sRPE is a useful strategy for assessing physical fatigue, while the psychomotor speed test assesses cerebral fatigue. Uniquely, tapering directly addresses the physical symptoms of OTS by recovering the body from accumulated fatigue, while periodization divides training programs into sequential, focused training periods. Ideally, an individual diagnosed with OTS is encouraged to assess both bodily and mental fatigue using each sRPE and psychomotor speed test, and employ tapering and periodization as a practical treatment option.

6.2. Implication

Understanding the factors and early warning signs of overtraining syndrome is essential for coaches and trainers to protect athletic performance and long-term health. By using assessment protocols, such as checklists, coaches can better monitor the athletes' stress levels and identify athletes who may be especially susceptible to overtraining. Closely tracking individual performance development, recovery patterns, and psychological well-being is essential for promoting sustainable athletic growth and maintaining a positive sporting experience. Training programs should include evaluating athletes' fatigue, mood, sleep quality, injury frequency, and motivation, particularly during and after regeneration or recovery periods,

when athletes' adaptation to training loads can be more accurately assessed. Adopting an athlete-centered approach to training is important. Training hard may seem to be the primary objective at first. However, long-term health and safety and athletes' ability to continue participating in training are ultimately more significant. Preventing chronic fatigue, burnout, and physical injury is essential not only for their athletic success but also for preserving their development and enjoyment of training. Therefore, monitoring their physical and mental health, balanced training-recovery programs, open communication between athletes and coaches, and education about the risks of consistent overtraining and overreaching should be essential components of modern athletic training systems.

7. Conclusion

In this paper, we addressed the research gap in overtraining syndrome specific to adolescent athletes, and ultimately discussed its physiological and psychological symptoms and suggested prevention strategies. Of importance, overtraining syndrome is prone to occur in athletes in the setting of excessive exercise not properly matched with recovery, and prolonged severe mood disruption. The paper largely focused on adolescent athletes, who we have discovered to be at higher risk of OTS than fully developed individuals. In addition, we suggested the optimal prevention-treatment option for athletes currently experiencing or exhibiting warning symptoms of OTS. Ultimately, the athletes are encouraged to communicate with coaches and clinicians and address warning signs, rather than risk their entire athletic lives over a temporary spark.

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